

606021



Test Booklet No.

ENGLISH

Test Booklet Code

06

NEET RANKERS

TEST SERIES-2025



Do not open this Test Booklet until you are asked to do so.

Important Instructions:

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars in ORIGINAL Copy carefully with blue/black ball point pen only.
- The test is of 3 hours 20 minutes duration and the Test Booklet contains 200 multiple-choice questions (four options with a single correct answer) from Physics, Chemistry and Biology (Botany and Zoology). 50 questions in each subject are divided into two Sections (A and B) as per details given below:
 - Section A shall consist of 35 (Thirty-five) Questions in each subject (Question Nos. 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B shall consist of 15 (Fifteen) questions in each subject (Question Nos. 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to attempt any 10 (Ten) questions out of 15 (Fifteen) in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
- Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- Use Blue/Black Ball Point Pen only for writing particulars on this page marking responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The CODE for this Booklet is 06. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is NOT permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Admit Card to the Invigilator.
- No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with name) the Attendance Sheet twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): _____

Roll Number: In figures _____

: In words _____

Centre of Examination (in Capitals): _____

Candidate's Signature: _____ Invigilator's Signature: _____

Facsimile Signature stamp of Centre Superintendent

Physics : Section-A (Q. No. 1 to 35)

1. The strength of the magnetic field at a distance r near a long straight current carrying wire is B .

The magnetic field at a distance $\frac{r}{2}$ will be:

- (1) $\frac{B}{2}$ (2) $\frac{B}{4}$
(3) $2B$ (4) $4B$

2. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The magnetism of magnet is due to the spin motion of electrons.

Reason (R): Dipole moment of spinning electron is smaller than that due to orbital motion around nucleus.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

3. Given below are two statements:

Statement I: The viscosity of a gas increases as the temperature rises.

Statement II: The viscosity of liquid decreases as the temperature rises.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I and incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

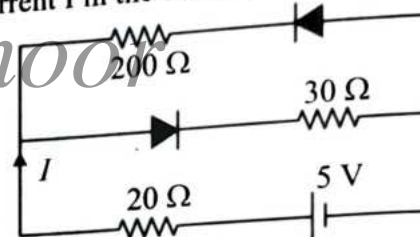
4. Two sources of light of the same frequency produce intensities I and $4I$ at a point P when used individually. If they are used together such that the light from them reach P with a phase difference of $\frac{2\pi}{3}$, then the intensity at P will be:

- (1) $2I$ (2) $3I$
(3) $4I$ (4) $5I$

5. In an AC circuit, a resistance R is connected in series with an inductance L . If phase angle between voltage and current be 45° , the value of inductive reactance will be:

- (1) $\frac{R}{4}$ (2) $\frac{R}{2}$
(3) R (4) $2R$

6. Current I in the circuit will be:



- (1) $\frac{5}{40}$ A (2) $\frac{5}{50}$ A
(3) $\frac{5}{10}$ A (4) $\frac{5}{20}$ A

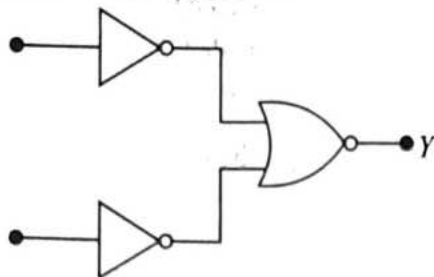
7. Two point charges q and $2q$ are placed some distance apart. If the electric field at the location of q be E , then that at the location of $2q$ will be:

- (1) $3E$ (2) $\frac{E}{2}$
(3) E (4) $2E$

8. In terms of basic units of mass (M), Length (L), time (T) and charge (Q), the dimensions of magnetic permeability of vacuum (μ_0) would be:

- (1) $[MLQ^{-2}]$ (2) $[LT^{-1}Q^{-1}]$
(3) $[ML^2T^{-1}Q^{-2}]$ (4) $[LTQ^{-1}]$

9. Which logic gate is represented by the following combination of logic gates?



- (1) OR
(2) NAND
(3) AND
(4) NOR
10. In an experiment four quantities a , b , c and d are measured with percentage error 1%, 2%, 3% and 4%, respectively. Quantity P is calculated as follows $P = \frac{a^3 b^2}{cd}$. Maximum percentage error

in P is:

- (1) 14%
(2) 10%
(3) 7%
(4) 4%
11. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

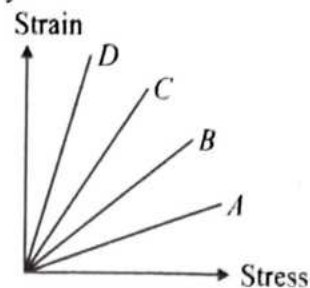
Assertion (A): In α -decay, atomic number of daughter nucleus reduces by a unit from the parents nucleus.

Reason (R): An α -particle carries four units of mass.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

12. Stress-strain curve for four metals are shown in figure. The maximum Young modulus of elasticity is for metal:



- (1) A
(2) B
(3) C
(4) D
13. A body of mass 40 kg resting on rough horizontal surface is subjected to a force P which is just enough to start the motion of the body. If $\mu_s = 0.5$, $\mu_k = 0.4$, $g = 10 \text{ ms}^{-2}$ and the force P is continuously applied on the body, then the acceleration of the body is:

- (1) zero
(2) 1 ms^{-2}
(3) 2 ms^{-2}
(4) 2.4 ms^{-2}
14. For photoelectric emission from certain metal, the cut-off frequency is ν . If radiation of frequency 2ν impinges on the metal plate, the maximum possible velocity of the emitted electron will be (m is the electron mass)

- (1) $\sqrt{\frac{h\nu}{2m}}$
(2) $\sqrt{\frac{h\nu}{m}}$
(3) $\sqrt{\frac{2h\nu}{m}}$
(4) $2\sqrt{\frac{h\nu}{m}}$

15. Two charged spheres of radii R_1 and R_2 having equal surface charge density. The ratio of their potential $\left(\frac{V_1}{V_2}\right)$ is:

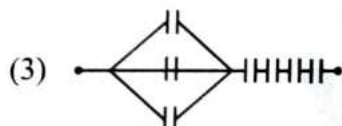
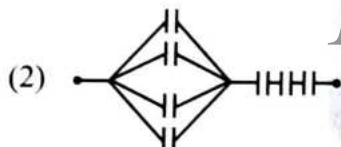
- (1) $\frac{R_1}{R_2}$
(2) $\frac{R_2}{R_1}$
(3) $\left(\frac{R_1}{R_2}\right)^2$
(4) $\left(\frac{R_2}{R_1}\right)^2$

16. A metal block is experiencing an atmospheric pressure of 10^5 Nm^{-2} . When the same block is placed in a vacuum chamber, the fractional change in its volume (the bulk modulus of metal is $1.25 \times 10^{11} \text{ Nm}^{-2}$) is:

- (1) 4×10^{-7}
- (2) 2×10^{-7}
- (3) 8×10^{-7}
- (4) 1×10^{-7}

17. Seven capacitors each of capacity $2\mu\text{F}$ are to be so connected to have a total capacity $\frac{10}{11}\mu\text{F}$.

Which will be the necessary figure as shown?



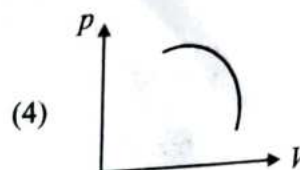
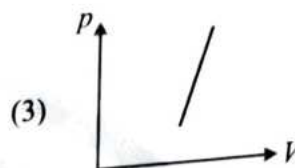
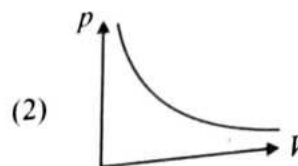
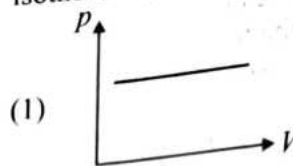
18. Which of the following statements are **CORRECT** regarding series LCR AC circuit?

Statement I: The current and voltage are in same phase at resonance.

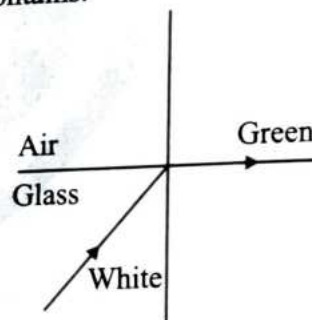
Statement II: The current at resonance does not depend on resistance.

- (1) I only
- (2) II only
- (3) I and II
- (4) Neither I nor II

19. Which of the p - V diagrams best represents an isothermal process?



20. White light is incident on the interface of glass and air as shown in the figure. If green light is just totally internally reflected, then the emerging ray in air contains.



- (1) yellow, orange, red
- (2) violet, indigo, blue
- (3) all colours
- (4) all colours except green

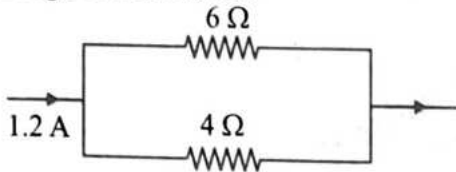
21. A uniform sphere of mass 500 g rolls without slipping on a plane surface so that its centre moves at a speed of 0.02 ms^{-1} . The total kinetic energy of rolling sphere would be (in J)

- (1) $1.4 \times 10^{-4} \text{ J}$
- (2) $0.75 \times 10^{-3} \text{ J}$
- (3) $5.75 \times 10^{-3} \text{ J}$
- (4) $4.9 \times 10^{-5} \text{ J}$

22. A bar magnet is hung by a thin cotton thread in a uniform horizontal magnetic field and is in equilibrium state. The energy required to rotate it by 60° is W . Now the torque required to keep the magnet in this new position is:

(1) $\frac{W}{\sqrt{3}}$ (2) $\sqrt{3}W$
 (3) $\frac{\sqrt{3}W}{2}$ (4) $\frac{2W}{\sqrt{3}}$

23. In the figure given below, the current passing through $6\ \Omega$ resistor is:



(1) 0.40 A (2) 0.48 A
 (3) 0.72 A (4) 0.80 A

24. The ratio of wavelength of the last line of Balmer series and the last line of Lyman series is:

(1) 2 (2) 1
 (3) 4 (4) 0.5

25. A proton and a deuteron, both having the same kinetic energy, enter perpendicularly into a uniform magnetic field B . For motion of proton and deuteron in circular path of radius R_p and R_d , respectively, the correct relation is:

(1) $R_d = \sqrt{2}R_p$ (2) $R_d = \frac{R_p}{\sqrt{2}}$
 (3) $R_d = R_p$ (4) $R_d = 2R_p$

26. A body of mass 5 kg is raised vertically to a height of 10 m by a force 170 N. The velocity of the body at this height will be: (take $g = 9.8\text{ m/s}^2$)

(1) 9.8 ms^{-1} (2) 15 ms^{-1}
 (3) 22 ms^{-1} (4) 37 ms^{-1}

27. The gravitational potential energy of a body of mass m at a distance r from the centre of the earth is U . What is the weight of the body at this distance?

(1) U (2) Ur
 (3) $\frac{U}{r}$ (4) $\frac{U}{2r}$

28. A radioactive nucleus ${}_{92}\text{X}^{235}$ decays to ${}_{91}\text{Y}^{231}$. Which of the following particles are emitted?

- (1) One alpha and one electron
 (2) Two deuterons and one positron
 (3) One alpha and one proton
 (4) One proton and four neutrons

29. A ball is projected horizontally from the top of a tower with a velocity v_0 . It will be moving at an angle of 60° with the horizontal after time.

(1) $\frac{v_0}{\sqrt{3}g}$ (2) $\frac{\sqrt{3}v_0}{g}$
 (3) $\frac{v_0}{g}$ (4) $\frac{v_0}{2g}$

30. The net work done by kinetic friction:

- (1) is always negative
 (2) is always zero
 (3) may be negative or positive
 (4) is always positive

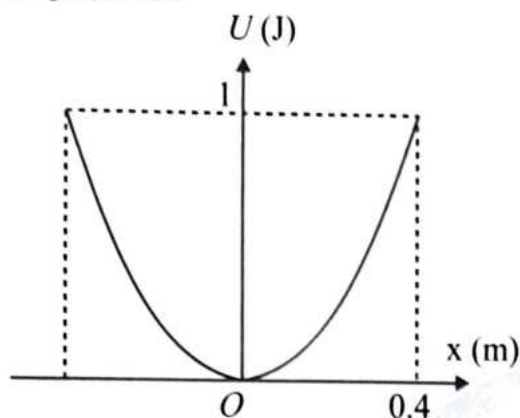
31. Out of the following options, which one can be used to produce a propagating electromagnetic wave?

- (1) A stationary charge
 (2) A chargeless particle
 (3) An accelerating charge
 (4) A charge moving at constant velocity

32. A point traversed $\left(\frac{3}{4}\right)^{\text{th}}$ of the circle of radius R in time t . The magnitude of the average velocity of the particle in this time interval is:

(1) $\frac{\pi R}{t}$ (2) $\frac{3\pi R}{2t}$
 (3) $\frac{R\sqrt{2}}{t}$ (4) $\frac{R}{\sqrt{2}t}$

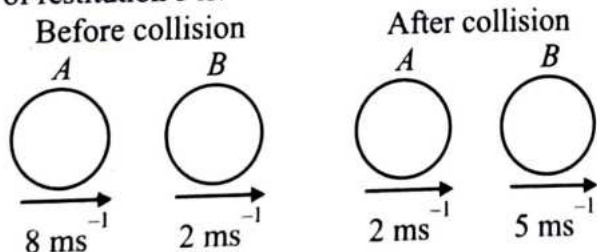
33. A particle of mass 2 kg moves in simple harmonic motion and its potential energy U varies with position x as shown. The period of oscillation of the particle is:



- (1) $\frac{2\pi}{5}$ s (2) $\frac{2\sqrt{2}\pi}{5}$
 (3) $\frac{\sqrt{2}\pi}{5}$ s (4) $\frac{4\pi}{5}$ s
34. The mean kinetic energy of one mole of gas per degree of freedom is:

- (1) $\frac{1}{2}kT$ (2) $\frac{3}{2}kT$
 (3) $\frac{3}{2}RT$ (4) $\frac{1}{2}RT$

35. The two diagrams show the situations before and after a collision between two spheres A and B of equal radii moving along the same straight line on a smooth horizontal surface. The coefficient of restitution e is:



- (1) $\frac{1}{3}$ (2) $\frac{1}{2}$
 (3) $\frac{2}{3}$ (4) $\frac{3}{4}$

Physics : Section-B (Q. No. 36 to 50)

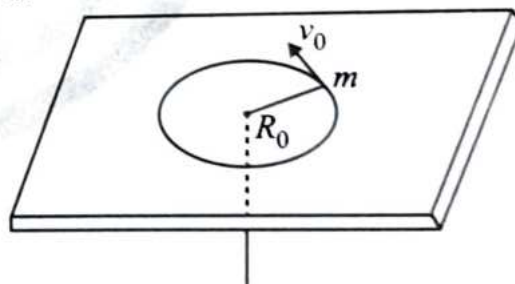
36. A rectangular film of liquid is extended from $(4 \text{ cm} \times 2 \text{ cm})$ to $(5 \text{ cm} \times 4 \text{ cm})$. If the work done is $3 \times 10^{-4} \text{ J}$, the value of the surface tension of the liquid is:

- (1) 0.250 Nm^{-1}
 (2) 0.125 Nm^{-1}
 (3) 0.2 Nm^{-1}
 (4) 8.0 Nm^{-1}

37. Which of the following cannot be polarised?

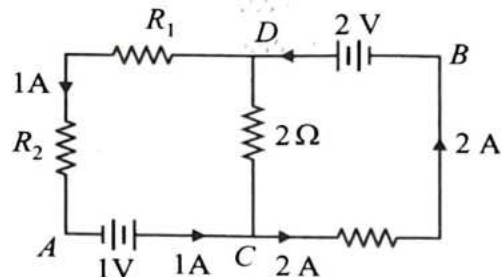
- (1) Radio waves
 (2) Ultraviolet rays
 (3) Infrared rays
 (4) Ultrasonic waves

38. A mass m moves in a circle on a smooth horizontal plane with velocity v_0 at a radius R_0 . The mass is attached to a string which passes through a smooth hole in the plane as shown in the figure. The tension in the string is increased gradually and finally m moves in a circle of radius $\frac{R_0}{2}$. The final value of the kinetic energy is:

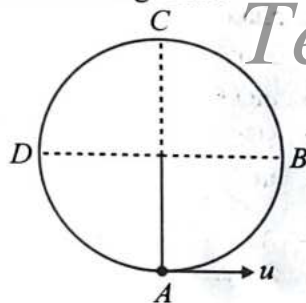


- (1) mv_0^2
 (2) $\frac{1}{4}mv_0^2$
 (3) $2mv_0^2$
 (4) $\frac{1}{2}mv_0^2$

39. In the circuit shown in the figure, if the potential at point A is taken to be zero, the potential at point B is:



- (1) +1 V
(2) -1 V
(3) +2 V
(4) -2 V
40. A particle is suspended from a string of length R . It is given a velocity $u = 3\sqrt{gR}$ at the bottom. Match the following Lists.



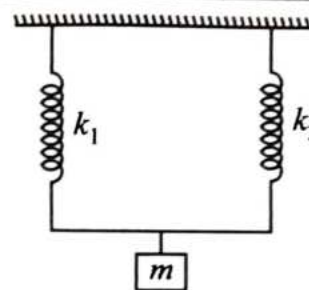
List-I

- (A) Velocity at B
(B) Velocity at C
(C) Tension in string at B
(D) Tension in string at C

List-II

- (I) $7mg$
(II) $\sqrt{5gR}$
(III) $\sqrt{7gR}$
(IV) $4mg$

- (1) A-III, B-II, C-I, D-IV
(2) A-IV, B-I, C-II, D-III
(3) A-III, B-II, C-IV, D-I
(4) A-II, B-III, C-I, D-IV
41. The spring constants of two springs of same length are k_1 and k_2 as shown in figure. if an object of mass m is suspended and set in vibration, the time period will be:



- (1) $2\pi\sqrt{\frac{mk_1}{k_2}}$ (2) $2\pi\sqrt{\frac{m}{k_1k_2}}$
(3) $2\pi\sqrt{\frac{m}{k_1 - k_2}}$ (4) $2\pi\sqrt{\frac{m}{k_1 + k_2}}$

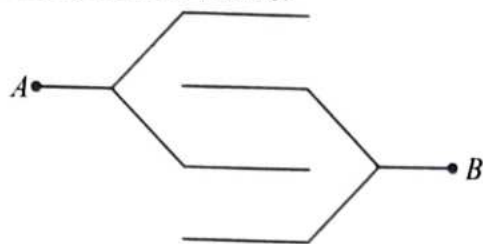
42. Positive charge Q is distributed uniformly over a circular ring of radius R . A point particle having a mass (m) and a negative charge $-q$ is placed on its axis at a distance x from the centre. Assuming $x < R$, find the time period of oscillation of the particle, if it is released from there [neglect gravity].

- (1) $\left[\frac{16\pi^3 \epsilon_0 R^3 m}{Qq} \right]^{\frac{1}{2}}$ (2) $\left[\frac{8\pi^3 \epsilon_0 R^3 m}{Qq} \right]^{\frac{1}{2}}$
(3) $\left[\frac{2\pi^3 \epsilon_0 R^3 m}{Qq} \right]^{\frac{1}{2}}$ (4) $\left[\frac{4\pi^3 \epsilon_0 R^3 m}{Qq} \right]^{\frac{1}{2}}$

43. Stationary waves are formed when:

- (1) two waves of equal amplitude and equal frequency travel along the same path with equal speed in opposite directions.
(2) two waves of equal wavelength and equal amplitude travel along the same path with equal speeds in same direction.
(3) two waves of equal wavelength and equal phase travel along the same path with equal speed.
(4) two waves of equal amplitude and equal speed travel along the same path in opposite directions.

44. Four plates of the same area of cross-section (A) are joined as shown in the figure. The distance between each plate is d . The equivalent capacity across A and B will be:



- (1) $\frac{2\epsilon_0 A}{d}$ (2) $\frac{3\epsilon_0 A}{d}$
(3) $\frac{3\epsilon_0 A}{2d}$ (4) $\frac{\epsilon_0 A}{d}$

45. Which of the following statements are **CORRECT** regarding spherical mirrors and real object?

Statement I: In case of a concave mirror, when the object is between F and P , the image formed is virtual and upright.

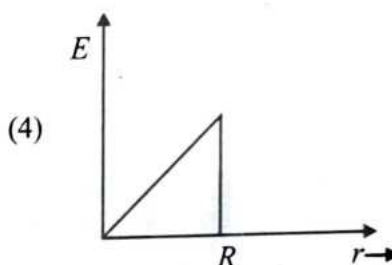
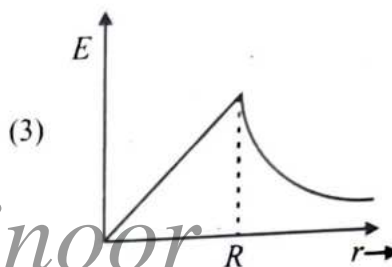
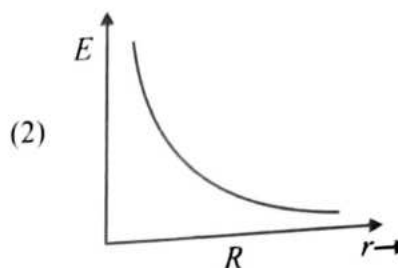
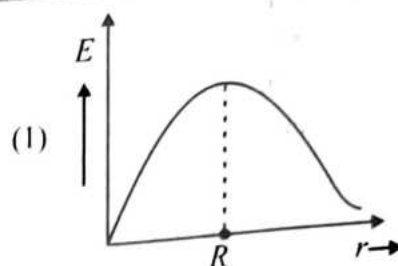
Statement II: In case of a convex mirror no matter where the object is, the image formed is always virtual and upright.

- (1) I only (2) II only
(3) I and II (4) Neither I nor II

46. The core used in a transformer and other electromagnetic devices is laminated so that:

- (1) ratio of voltage in the primary and secondary may be increased.
(2) energy loss due to eddy currents may be minimised.
(3) the weight of the transformer may be reduced.
(4) residual magnetism in the core may be reduced.

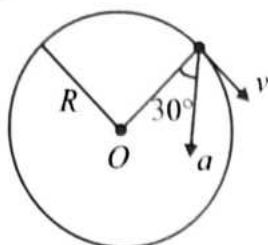
47. In a uniformly charged non-conducting sphere of total charge Q and radius R , the electric field E is plotted as function of distance from the centre. The graph which would correspond to the above will be:



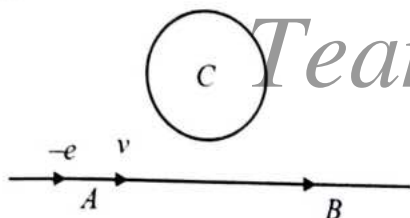
48. A solid shell loses half, its weight in water. Relative density of shell is 5.0. What fraction of its volume is hollow?

- (1) $\frac{3}{5}$
(2) $\frac{2}{5}$
(3) $\frac{1}{5}$
(4) $\frac{4}{5}$

49. In the given figure, $a = 15 \text{ m/s}^2$ represents the total acceleration of a particle moving in the clockwise direction in a circle of radius $R = 2.5 \text{ m}$ at a given instant of time. The speed of the particle is:



- (1) 4.5 m/s (2) 3.7 m/s
(3) 5.7 m/s (4) 6.7 m/s
50. An electron moves along the line AB which lies in the same plane as a circular loop of conducting wire as shown in figure. What will be the direction of the current induced (if any) in the loop?



- (1) No current will be induced
(2) The current will be clockwise
(3) The current will be anti-clockwise
(4) The current will change direction as the electron passes through

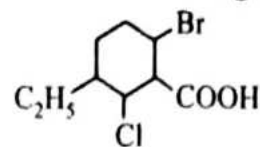
Chemistry : Section-A (Q. No. 51 to 85)

51. What happens when SO_2 is dissolved in water?
- (1) Sulphurous acid is formed.
(2) SO_3 is formed.
(3) Liquid SO_2 is formed.
(4) None of these.
52. 6.02×10^{20} molecules of urea are present in 100 mL of its solution. The concentration of solution is; (Given $N_A = 6.02 \times 10^{23}$)
- (1) 0.05 M (2) 0.01 M
(3) 0.02 M (4) 0.04 M

53. Neutral oxide among the following is;

- (1) NO (2) SO_2
(3) N_2O_3 (4) SO_3

54. The correct IUPAC name of given compound is;



- (1) 2-Bromo-6-chloro-5-ethyl-cyclohexan-1-oic acid
(2) 6-Bromo-2-chloro-5-ethyl-cyclo-hexan-1-oic acid
(3) 6-Bromo-2-chloro-3-ethyl-cyclohexan-1-oic acid
(4) 2-Bromo-5-ethyl-6-chloro-cyclohexan-1-oic acid

55. Which of the following is **not** an example of redox reactions?

- (1) $2\text{MnO}_4^- + \text{F}^- + \text{H}_2\text{O} \rightarrow \text{IO}_3^- + 2\text{MnO}_2 + 2\text{OH}^-$
(2) $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HClO}$
(3) $2\text{CrO}_4^{2-} + 2\text{H}^+ \rightarrow \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$
(4) $\text{MnO}_4^- + 8\text{H}^+ + 5\text{Ag} \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} + 5\text{Ag}^+$

56. Which of the following is **incorrect**?

- (1) $E_{\text{cell}}^\circ = -\frac{2.303RT}{nF} \log K_c$
(2) $\Delta G^\circ = -nFE_{\text{cell}}^\circ$
(3) $\Delta G^\circ = 2.303 RT \log K_c$
(4) Both (1) and (3)

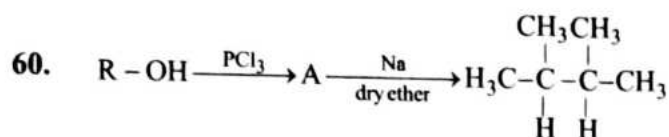
57. The electronic configuration of Gadolinium (Atomic number = 64) is;

- (1) $[\text{Xe}]4f^85d^96s^2$ (2) $[\text{Xe}]4f^95d^16s^2$
(3) $[\text{Xe}]4f^85d^56s^2$ (4) $[\text{Xe}]4f^85d^26s^2$

58. The correct order of energy for H atoms;

- (1) $1s < 2s < 2p < 3s$
(2) $1s < 2s = 2p = 3s$
(3) $1s = 2s = 2p = 3s$
(4) $1s < 2s = 2p < 3s$

59. Among the following pairs of compound which show negative deviation from Raoult's law is/are;
- (1) H_2O , HCl
 - (2) CHCl_3 , CH_3COCH_3
 - (3) $\text{C}_6\text{H}_5\text{OH}$, $\text{C}_6\text{H}_5\text{NH}_2$
 - (4) All of these



The value of R in above reaction is

- (1) $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-$
- (2) $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{H}}{\text{C}}}-$
- (3) CH_3-CH_2-
- (4) $\text{H}_3\text{C}-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{CH}_2-$

61. In a multiple bond, shifting of electron under the influence of a reagent is called;
- (1) Inductive effect
 - (2) Mesomeric effect
 - (3) Electromeric effect
 - (4) Trans effect

62. Given below are two statements:

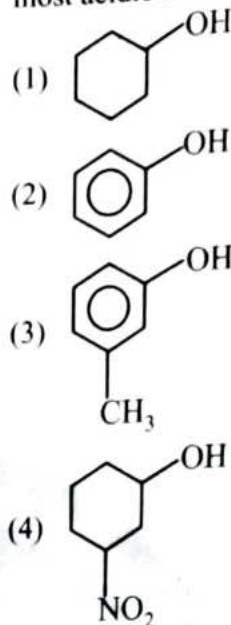
Statement I: In cyclic process, ΔU and ΔH is zero.

Statement II: When initial and final state of a system in a process are same, the process is called a cyclic process.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

63. Which one of the following compounds has the most acidic nature?



64. Given below are two statements:

Statement I: For most ionic compounds, $\Delta_{\text{sol}}H^\circ$ is positive at 298 K and dissolution process is endothermic.

Statement II: Born-Haber cycle is used to calculate the lattice enthalpy of an ionic compound indirectly.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

65. What is amount of CaCl_2 ($i = 3$) dissolved in 4 litre of water such that its osmotic pressure is 1 atm at 127°C ? [Molar mass of $\text{CaCl}_2 = 111\text{g mol}^{-1}$]

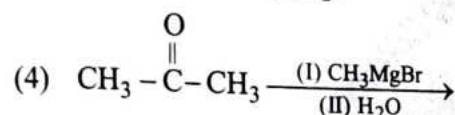
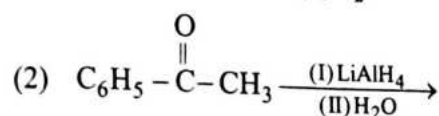
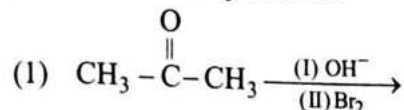
- (1) 8 g
- (2) 4.5 g
- (3) 6 g
- (4) 2 g



66. Nessler's reagent is used to detect

- (1) CrO_4^{2-} (2) PO_4^{3-}
(3) MnO_4^{2-} (4) NH_4^+

67. Which one of the following reaction would produce secondary alcohol?



68. Crystal field stabilization energy for high spin d^4 octahedral complex is;

- (1) $-1.8 \Delta_o$
(2) $-1.6 \Delta_o$
(3) $-1.2 \Delta_o$
(4) $-0.6 \Delta_o$

69. Given below are two statements:

Statement I: Lactose is a disaccharide made up of glucose and galactose unit.

Statement II: Such a linkage between two monosaccharide units through oxygen atom is called glycosidic linkage.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

70. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Tertiary amines do not react with benzenesulphonyl chloride.

Reason R: The reaction of benzenesulphonyl chloride with primary amine yields N, N-Dialkyl-benzenesulphonamide.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

71. The energies E_1 and E_2 of two radiations are 75 eV and 300 eV respectively. The relations between their wavelengths, i.e. λ_1 and λ_2 will be;

- (1) $\lambda_1 = 4\lambda_2$ (2) $\lambda_1 = 2\lambda_2$
(3) $\lambda_1 = \frac{1}{2}\lambda_2$ (4) $\lambda_1 = \lambda_2$

72. $2\text{N}_2\text{O}_5 \rightleftharpoons 4\text{NO}_2 + \text{O}_2$, for given reaction which of the following is incorrect about rate of the reaction?

- (1) $-\frac{d[\text{N}_2\text{O}_5]}{dt} = 2 \frac{d[\text{O}_2]}{dt}$
(2) $-\frac{1}{2} \frac{d[\text{N}_2\text{O}_5]}{dt} = \frac{d[\text{O}_2]}{dt}$
(3) $\frac{d[\text{NO}_2]}{dt} = 4 \frac{d[\text{O}_2]}{dt}$
(4) $-\frac{d[\text{N}_2\text{O}_5]}{dt} = \frac{d[\text{NO}_2]}{dt} = 4 \frac{d[\text{O}_2]}{dt}$

73. The rate of reaction doubles by increase of 10°C temperature. If there is an increase of 50°C in temperature, how many times rate of reaction would increase?

- (1) 64 times
(2) 32 times
(3) 10 times
(4) 24 times

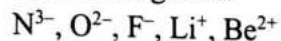
74. Match List-I with List-II.

List-I	List-II
A. $R-Mg-X$	I. 1° alcohol
B. $CH_3CH=CH_2 \xrightarrow[(ii) OH^-]{(i) B_2H_6/THF}$	II. Iodoform test
C. $(CH_3)_2CO \xrightarrow{NaOI}$	III. Grignard reagent
D. $CH_3CHO \xrightarrow[\Delta]{dil. NaOH}$	IV. Aldol condensation

Choose the correct answer from the options given below:

- (1) A-IV, B-I, C-III, D-II
 (2) A-III, B-I, C-II, D-IV
 (3) A-I, B-III, C-II, D-IV
 (4) A-II, B-I, C-III, D-IV

75. What is the correct decreasing order of ionic radii of following ions:



- (1) $N^{3-} > O^{2-} > F^- > Be^{2+} > Li^+$
 (2) $N^{3-} > O^{2-} > F^- > Li^+ > Be^{2+}$
 (3) $N^{3-} > O^{2-} > Be^{2+} > Li^+ > F^-$
 (4) $Li^+ > F^- > O^{2-} > Be^{2+} > N^{3-}$

76. The given orbital which have electron density along the axis is;

- (1) d_{z^2}, d_{xy} (2) $d_{x^2-y^2}, d_{yz}$
 (3) $d_{x^2-y^2}, p_x$ (4) d_{xy}, d_{yz}

77. Which of the following does not follow Huckel rule?



- (1) II only (2) I and II only
 (3) II and III only (4) I, II and III only

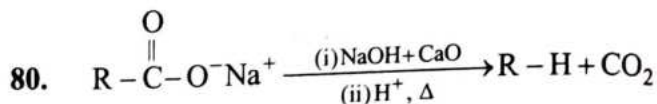
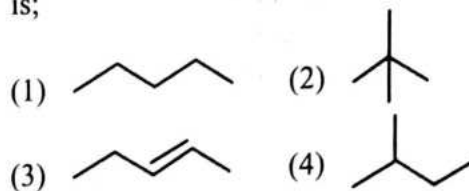
78. For the reaction:



The unit of K_p for the reactions is;

- (1) atm^{-1} (2) atm^{-2}
 (3) atm^2 (4) atm

79. A hydrocarbon C_5H_{12} does not react with chlorine in dark but gives only one monochloro compound in bright sunlight. Then hydrocarbon is;



This decarboxylation reaction occurs through the generation of;

- (1) $R^\oplus$ (2) $\dot{R}$
 (3) $R^\ominus$ (4) All of these

81. At which of the following temperatures the reaction $C(\text{graphite}) + CO_2(g) \rightarrow 2CO(g)$ will occur spontaneously if $\Delta H = 170 \text{ kJ mol}^{-1}$ and $\Delta S = 170 \text{ JK}^{-1} \text{ mol}^{-1}$?

- (1) 710 K (2) 1110 K
 (3) 910 K (4) 510 K

82. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R: **Assertion A:** Nodal plane of p_x atomic orbital is yz plane.

Reason R: In p_x atomic orbital electron density is zero in the yz plane.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

83. A first order reaction has a specific reaction rate of 10^{-1} s^{-1} . How much time will it take for 100 g of its reactant to be reduced to 25 g?

- (1) 14 sec (2) 28 sec
 (3) 10 sec (4) 39 sec

84. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: One molal solution prepared at 20°C will retain the same molality at 100°C, provided there is no loss of solute or solvent on heating.

Reason R: Molality is independent of temperature.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

85. Bromobenzene reacts with Mg in the presence of dry ether to give a compound (A) which further reacts with CH_3OH and give compound (B). Identify (B) is;

- (1) phenol
- (2) benzene
- (3) methyl benzene
- (4) phenyl methyl ether

Chemistry : Section-B (Q. No. 86 to 100)

86. $\text{Zn}|\text{Zn}^{2+}(0.05 \text{ M})||\text{Fe}^{2+}(0.005 \text{ M})|\text{Fe}$.
The potential difference of the above cell is 0.0885 V. Equilibrium constant for the cell reaction is;

- (1) 10^2
- (2) 10^4
- (3) 10
- (4) 10^8

87. Which one is diamagnetic and has the largest bond length?

- (1) C_2^{2-}
- (2) O_2
- (3) O_2^{2-}
- (4) N_2^{2-}

88. A lithium atom has;

- (1) 3 protons, 3 electrons and 4 neutrons
- (2) 3 protons, 4 electrons and 3 neutrons
- (3) 3 protons, 3 electrons and 7 neutrons
- (4) 3 protons, 7 electrons and 7 neutrons

89. Match List-I with List-II.

List-I		List-II	
(Reaction name)		(Reagent)	
(A)	Wurtz reaction	(I)	$\text{H}_2/\text{Pd}-\text{BaSO}_4$
(B)	Clemmensen's reduction	(II)	$\text{N}_2\text{H}_4 / \text{KOH} / \text{HOCH}_2\text{CH}_2\text{OH}$
(C)	Rosenmund's reduction	(III)	$\text{Na}/\text{dry ether}$
(D)	Wolff-Kishner reduction	(IV)	$\text{Zn}-\text{Hg}/\text{conc. HCl}$

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-III, B-IV, C-II, D-I
- (3) A-II, B-I, C-IV, D-III
- (4) A-IV, B-III, C-II, D-I

90. $\text{Co}|\text{Co}^{2+}(\text{C}_2)||\text{Co}^{2+}(\text{C}_1)|\text{Co}$, this cell reaction will be spontaneous only when;

- (1) $\text{C}_2 > \text{C}_1$
- (2) $\text{C}_1 > \text{C}_2$
- (3) $\text{C}_1 = \text{C}_2$
- (4) Both (2) and (3)

91. The correct order of first ionization enthalpy is;

- (1) $\text{Fe} > \text{Co} > \text{Ni} > \text{Mn}$
- (2) $\text{Co} > \text{Fe} > \text{Ni} > \text{Mn}$
- (3) $\text{Fe} > \text{Co} > \text{Mn} > \text{Ni}$
- (4) $\text{Co} > \text{Fe} > \text{Mn} > \text{Ni}$

92. Given below are two statements:

Statement I: 3 molecules of ethyne on passing through red hot iron tube at 873 K undergoes cyclic polymerization.

Statement II: Benzene is formed as the product during cyclic polymerization of ethyne.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

93. Aryl halides are less reactive towards nucleophilic substitution reactions as compared to alkyl halides due to;

- (1) formation of a less stable carbonium ion in aryl halides.
- (2) resonance stabilization in aryl halides.
- (3) presence of double bonds in alkyl halides.
- (4) inductive effect in aryl halides.

94. Number of unpaired electrons and hybridisation of $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is;

- (1) 3, d^2sp^3
- (2) 2, d^2sp^3
- (3) 3, sp^3d^2
- (4) 2, sp^3d^2

95. Match List-I with List-II.

List-I (Compounds)	List-II (Shape)
(A) SF_4	(I) Tetrahedral
(B) BrF_3	(II) Pyramidal
(C) BrO_3^-	(III) See-saw shaped
(D) NH_4^+	(IV) T-shaped

Choose the correct answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-III, B-IV, C-II, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-I, B-IV, C-III, D-II

96. Oxidation state of N in KNO_3 is;

- (1) -1
- (2) +3
- (3) +5
- (4) -3

97. Examine the following statements regarding the dehydrohalogenation of alkyl halides in the synthesis of alkenes;

- A. Alcoholic KOH is used for dehydrohalogenation.
- B. Rate of dehydrohalogenation for halogens: Iodide > Bromide > Chloride.
- C. Rate of dehydrohalogenation of alkyl halides: Primary > Secondary > Tertiary.

Choose the most appropriate incorrect answer from the options given below:

- (1) B and C only
- (2) B only
- (3) A and B only
- (4) C only

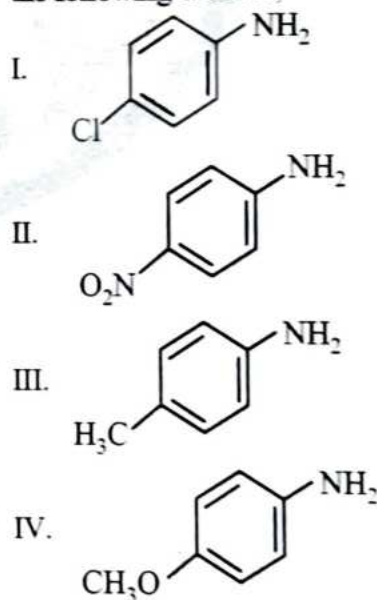
98. Match List-I with List-II.

List-I (Nodes)	List-II (Number)
(A) Radial nodes of 4s atomic orbital	(I) 1
(B) Angular nodes of $4d_{yz}$ atomic orbital	(II) 3
(C) Angular nodes + radial nodes of $5d_{xy}$ atomic orbital	(III) 2
(D) Angular nodes of 2p atomic orbital	(IV) 4

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-II, B-III, C-I, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-II, B-I, C-IV, D-III

99. The correct order of increasing basic nature of the following bases is;



- (1) II < IV < III < I
- (2) II < I < III < IV
- (3) I < III < IV < II
- (4) II < I < IV < III

100. For a particular reaction, if ΔG° is positive then;
- (1) $K_{eq} = 1$
 - (2) $K_{eq} < 100$
 - (3) $K_{eq} > 1$
 - (4) $K_{eq} < 1$

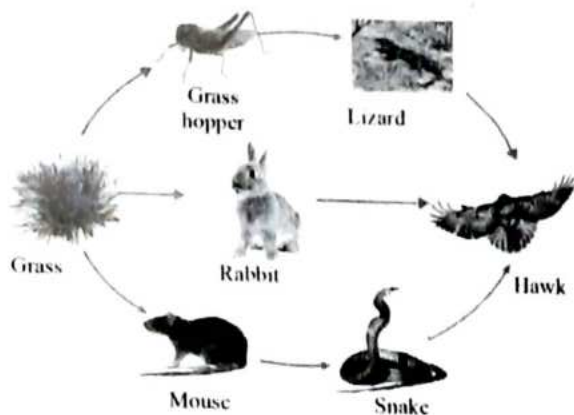
Botany : Section-A (Q. No. 101 to 135)

101. Choose the **correct** sequence of layers of anther from outer to inner side.
- (1) Epidermis $\rightarrow$ Tapetum $\rightarrow$ Middle layers $\rightarrow$ Endothecium
 - (2) Tapetum $\rightarrow$ Endothecium $\rightarrow$ Epidermis $\rightarrow$ Middle layers
 - (3) Epidermis $\rightarrow$ Endothecium $\rightarrow$ Middle layers $\rightarrow$ Tapetum
 - (4) Endothecium $\rightarrow$ Epidermis $\rightarrow$ Middle layers $\rightarrow$ Tapetum
102. Golden rice is a transgenic crop of the future with which of the following improved traits?
- (1) High lysine (essential amino acid) content
 - (2) Insect resistance
 - (3) High protein content
 - (4) High vitamin-A content
103. Match the columns and select the correct option.
- | Column-I | | Column-II | |
|----------|---------------|-----------|---------------------|
| (a) | Free living | (i) | <i>Glomus</i> |
| (b) | Mycorrhiza | (ii) | <i>Azotobacter</i> |
| (c) | Cyanobacteria | (iii) | <i>Rhizobium</i> |
| (d) | Symbiotic | (iv) | <i>Oscillatoria</i> |
- (1) (a)-(iii) (b)-(i) (c)-(ii) (d)-(iv)
 - (2) (a)-(iv) (b)-(ii) (c)-(iii) (d)-(i)
 - (3) (a)-(ii) (b)-(i) (c)-(iv) (d)-(iii)
 - (4) (a)-(iv) (b)-(iii) (c)-(ii) (d)-(i)
104. Choose the **incorrect** statement w.r.t. transcription.
- (1) It is the process of copying genetic information from one strand of the DNA into RNA.
 - (2) Adenine pairs with Uracil instead of Thymine.

- (3) An initiation factor (σ factor) transiently binds to RNA polymerase to initiate transcription.
- (4) It takes place in $3' \rightarrow 5'$ direction.

105. Negatively charged DNA is wrapped around histone octamer to form a:
- (1) Nucleotide.
 - (2) Nucleosome.
 - (3) Euchromatin.
 - (4) Heterochromatin.
106. In which of the following methods, are high-velocity gold or tungsten particles coated with recombinant DNA being bombarded on host cells?
- (1) Polymerase Chain Reaction
 - (2) Biolistics
 - (3) Elution
 - (4) Microinjection
107. In eukaryotes, rRNAs are transcribed by:
- (1) RNA polymerase I and III.
 - (2) RNA polymerase III only.
 - (3) RNA polymerase I and II.
 - (4) RNA polymerase I only.
108. In the offsprings of the F_2 generation of a dihybrid cross, how many different combinations of genotypes can you expect to obtain?
- (1) 6
 - (2) 4
 - (3) 9
 - (4) 16
109. Vegetative cell of pollen grain is:
- (1) smaller than the generative cell.
 - (2) having an abundant food reserve.
 - (3) spindle shaped with dense cytoplasm.
 - (4) with a small spherical nucleus.
110. Which among the following products are produced from a bacterium?
- (a) Citric acid
 - (b) Acetic acid
 - (c) Butyric acid
 - (d) Cyclosporin A
 - (e) Statin
- (1) (a), (c) and (e)
 - (2) (b) and (c)
 - (3) (c) and (d)
 - (4) (a), (b) and (d)

111. In the given food web, an increase in the population of hawks will not result in



- (1) Decrease in the population of rabbits and snakes
- (2) Increase in the population of producers
- (3) Decrease in the population of lizards
- (4) Decrease in the population of grasshoppers.

112. The restriction enzyme responsible for the cleavage of the following sequence is:

5'- G- A- A- T- T- C- 3'

3'- C- T- T- A- A- G- 5'

- (1) *EcoR* I. (2) *Hind* II.
- (3) *Bam*H I. (4) *EcoR* II.

113. Nucleolus and the nuclear envelope reappear at which stage of mitosis?

- (1) Prophase (2) Metaphase
- (3) Anaphase (4) Telophase

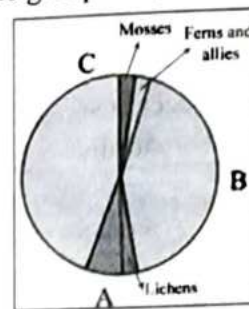
114. A 'new' variety of rice was patented by a foreign company, though such varieties have been present in India for a long time. This is related to:

- (1) basmati rice.
- (2) Bt rice.
- (3) golden rice.
- (4) All of these

115. How many turns of the Calvin cycle are required to synthesize one molecule of glucose?

- (1) Two (2) Four
- (3) Six (4) Eight

116. Identify the groups- A, B and C



- (1) A → Angiosperms, B → Algae, C → Molluscs
- (2) A → Angiosperms, B → Algae, C → Echinoderms
- (3) A → Algae, B → Angiosperms, C → Echinoderms
- (4) A → Algae, B → Angiosperms, C → Fungi

117. Select the **incorrect** statement.

- (1) The detritus food chain may be connected with the grazing food chain at some levels.
- (2) In an aquatic ecosystem, GFC is the major conduit for energy flow.
- (3) In a terrestrial ecosystem, a much larger fraction of energy flows through the GFC than through the detritus food chain (DFC).
- (4) Based on the source of their nutrition or food, organisms occupy a specific place in the food chain that is known as their trophic level.

118. Which of the following are examples of *in-situ* conservation methods?

- (a) National parks (b) Zoological parks
- (c) Seed banks (d) Sacred groves

Choose the correct answer from the options given below:

- (1) (a) and (d) only
- (2) (a), (b) and (c) only
- (3) (b), (c) and (d) only
- (4) All of the above



119. Choose the **incorrect** statement from the following w.r.t 'temperature' as a factor affecting photosynthesis.

- (1) The dark reactions being enzymatic are temperature controlled.
- (2) Light reactions are temperature sensitive but are affected to a much lesser extent.
- (3) The C_4 plants respond to higher temperatures and show a higher rate of photosynthesis.
- (4) Temperate plants have a higher temperature optimum than the plants adapted to tropical climates.

120. The margin of the thalamus grows upward enclosing the ovary completely and getting fused with it, the other parts of the flower arise above the ovary. This is **true** for:

- (1) Rose. (2) Guava.
- (3) China rose. (4) Peach.

121. Which of the following are components of cyclic photophosphorylation?

- (a) Photosystem II (b) Photosystem I
- (c) ATP synthase (d) Chlorophyll P700
- (e) NADPH reductase

Choose the correct answer from the options given below:

- (1) (a), (b), (d) only (2) (b), (c), (e) only
- (3) (b), (c), (d) only (4) (a), (c), (d) only

122. Match List-I with List-II:

List-I	List-II
A. Tobacco budworm	I. Nematode
B. Beetles	II. Dipterans
C. Mosquitoes	III. Lepidopterans
D. <i>Meloidegryne incognita</i>	IV. Coleopterans

Choose the **correct** answer from the options given below:

- (1) A- IV, B-III, C-II, D-I
- (2) A-III, B-I, C-IV, D-II
- (3) A-II, B-III, C-I, D- IV
- (4) A-III, B- IV, C-II, D-I

123. Match List-I with List-II.

List-I (Characters)	List-II (Recessive trait)
(A) Flower colour	(I) White
(B) Seed colour	(II) Green
(C) Seed shape	(III) Wrinkled
(D) Flower position	(IV) Terminal

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-II, B-I, C-III, D-IV
- (4) A-I, B-II, C-III, D-IV

124. The plant growth regulator that could fit either of the groups, promoter as well inhibitor, but it is largely an inhibitor of growth activities, is:

- (1) ABA.
- (2) ethylene.
- (3) gibberellin.
- (4) auxin.

125. Which of the following plant is bisexual, syncarpous, superior ovary, actinomorphic, parietal placentation, tap root system, variation in the length of filaments within a flower?

- (1) Mustard (2) Sweet pea
- (3) Soyabean (4) Moong

126. The group Chrysophytes includes:

- (1) diatoms and desmids (golden algae).
- (2) euglenoids.
- (3) dinoflagellates.
- (4) slime moulds.

127. Match List-I with List-II.

List-I	List-II
(A) Interphase	(I) Centriole duplication in the cytoplasm
(B) S phase	(II) Most dramatic phase of cell cycle
(C) M phase	(III) Resting phase of cell cycle
(D) G ₁ phase	(IV) Interval between mitosis and DNA replication

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
 (2) A-IV, B-II, C-III, D-I
 (3) A-III, B-I, C-II, D-IV
 (4) A-IV, B-III, C-II, D-I

128. Evolutionarily, the first **terrestrial** plants to possess vascular tissues are:

- (1) algae. (2) liverworts.
 (3) pteridophytes. (4) mosses.

129. The peripheral doublets of cilium or the flagellum are interconnected by:

- (1) axoneme. (2) central sheath.
 (3) linkers. (4) radial spoke.

130. Plants follow different pathways in response to the environment or phases of life to form different kinds of structures. This ability is called:

- (1) dedifferentiation. (2) plasticity.
 (3) differentiation. (4) development.

131. In general, viruses that infect plants have:

- (1) ss DNA (2) ds RNA
 (3) ds DNA (4) ss RNA

132. Dissolution of the synaptonemal complex occurs in:

- (1) diplotene. (2) pachytene.
 (3) zygotene. (4) diakinesis.

133. The Earth Summit held in Rio de Janeiro in 1992 was called

- (1) To reduce CO₂ emissions and global warming
 (2) For conservation of biodiversity and sustainable utilization of its benefits
 (3) To assess threat posed to native species by invasive weed species
 (4) For immediate steps to discontinue use of CFCs that were damaging the ozone layer

134. Match List-I with List-II.

List-I	List-II
(A) Sigma factor	(I) Aminoacylation of tRNA
(B) Transcription	(II) Termination factor
(C) Translation	(III) Initiation factor
(D) Rho-factor	(IV) DNA dependent RNA polymerase

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
 (2) A-II, B-IV, C-III, D-I
 (3) A-II, B-III, C-IV, D-I
 (4) A-IV, B-II, C-III, D-I

135. Match the column-I and column-II and find out the correct combination:

Column-I	Column-II
(a) Agar	(i) <i>Gelidium, Gracilaria</i>
(b) Algin	(ii) Brown algae
(c) Carrageen	(iii) Red algae
(d) <i>Chlorella</i>	(iv) Used as food supplements by space travellers

Choose the correct combination

- (1) (a)-(i) (b)-(ii) (c)-(iii) (d)-(iv)
 (2) (a)-(iv) (b)-(iii) (c)-(ii) (d)-(i)
 (3) (a)-(ii) (b)-(i) (c)-(iii) (d)-(iv)
 (4) (a)-(iii) (b)-(ii) (c)-(i) (d)-(iv)


Botany : Section-B (Q. No. 136 to 150)

136. Match List-I with List-II w.r.t dicot root.

List-I	List-II
(A) Conjunctive tissue	(I) Water-impermeable, waxy material made of suberin
(B) Casparian strips	(II) The outermost layer
(C) Pericycle	(III) The parenchymatous cells which lie between the xylem and the phloem
(D) Epiblema	(IV) Initiation of lateral roots and vascular cambium during the secondary growth

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-IV, B-II, C-III, D-I
- (3) A-III, B-I, C-IV, D-II
- (4) A-IV, B-III, C-II, D-I

137. Find out the **incorrect** statement from the following w.r.t. respiratory quotient (RQ).

- (1) When carbohydrates are used as substrate and are completely oxidised, the RQ will be 1.
- (2) When proteins are used as a substrate in respiration, the RQ is about 0.9.
- (3) When tripalmitin is used as a substrate in respiration, the RQ is 0.7.
- (4) Pure proteins or fats are always preferred as respiratory substrates.

138. Choose the **incorrectly** matched pair from the following w.r.t dorsiventral leaf.

- (1) Stomata - generally more on the abaxial surface.
- (2) Mesophyll - not differentiated into palisade and spongy parenchyma.
- (3) Venation - reticulate.
- (4) Bundle sheath cells - surround the vascular bundles.

139. Given below are two statements:

Statement I: 2'-OH group present at every nucleotide in RNA is a reactive group and makes RNA easily degradable.

Statement II: RNA is single stranded and known to be catalytic, hence stable.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

140. Match List-I with List-II.

List-I	List-II
(A) Double membrane	(I) Vacuole
(B) Single membrane	(II) Mitochondria
(C) Non-membranous	(III) Mesosome
(D) Extension of plasma membrane	(IV) Ribosome

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-II, C-III, D-IV
- (3) A-II, B-I, C-IV, D-III
- (4) A-III, B-II, C-I, D-IV

141. Arrange the following diagrams according to the correct order of developmental stages.



- (1) $A \rightarrow C \rightarrow B$ (2) $A \rightarrow B \rightarrow C$
(3) $C \rightarrow B \rightarrow A$ (4) $C \rightarrow A \rightarrow B$

142. Anaphase stage is characterised by:

- A. attachment of spindle fibres to the kinetochores of chromosomes.
B. splitting of centromeres and separation of chromatids.
C. moving of chromosomes to spindle equator and getting aligned along metaphase plate through spindle fibres to both poles.
D. moving of chromatids to the opposite poles.

Choose the correct option.

- (1) A and C (2) C and D
(3) B and D (4) A and B

143. Study the following table.

Cell organelle	Discoverer	Function
(a) Ribosomes	Palade	Protein synthesis
(b) Golgi complex	Golgi	DNA synthesis
(c) Cell	Flemming	Fat metabolism
(d) Nucleus	Robert Brown	Stores genetic information.

Which two show the correct combination?

- (1) (a) and (b) (2) (b) and (c)
(3) (c) and (d) (4) (a) and (d)

144. All of the following fungi reproduce by asexual spores known as conidia, **except**:

- (1) *Alternaria* (2) *Colletotrichum*
(3) *Agaricus* (4) *Trichoderma*

145. Match List-I with List-II.

List-I	List-II
(A) Molecular oxygen	(I) Leads to formation of α -ketoglutaric acid from citrate.
(B) Electron carrier	(II) Terminal electron acceptor in ETS
(C) Pyruvate dehydrogenase	(III) Cytochrome c
(D) Decarboxylation	(IV) Converts pyruvate to Acetyl CoA

Choose the correct answer from the options given below:

- (1) A-IV, B-II, C-III, D-I
(2) A-II, B-III, C-IV, D-I
(3) A-I, B-III, C-IV, D-II
(4) A-II, B-IV, C-III, D-I

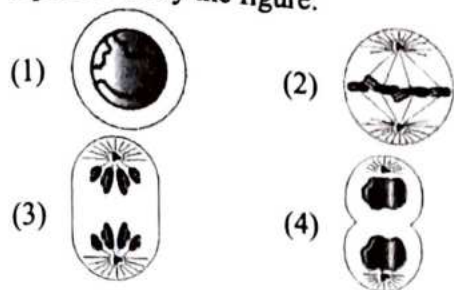
146. Match List-I with List-II.

List-I	List-II
(A) Ethylene	(I) Apical dominance
(B) Cytokinin	(II) Bolting
(C) Gibberellins	(III) Overcome apical dominance
(D) Auxin	(IV) Breaks seed and bud dormancy

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-II, D-IV
(4) A-IV, B-II, C-III, D-I

147. In mitosis, the stage in which daughter chromosomes of the future daughter nuclei begin their migration towards the two opposite poles is represented by the figure:



148. Match List-I with List-II.

List-I	List-II
(A) Pulvinus	(I) Swollen leaf base
(B) Root cap	(II) Green expanded part of leaf
(C) Lamina	(III) Thimble like structure
(D) Petiole	(IV) Stalk that attaches the leaf blade to stem

Choose the correct answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
 (2) A-III, B-II, C-I, D-IV
 (3) A-I, B-III, C-II, D-IV
 (4) A-II, B-I, C-III, D-IV
149. A single-stranded nucleic acid tagged with a radioactive molecule is called:
 (1) plasmid. (2) probe.
 (3) vector. (4) selectable marker.
150. **Statement-I:** The TCA cycle starts with the condensation of acetyl group with oxaloacetic acid (OAA) and water to yield citric acid.
Statement-II: Oxaloacetate is a 5C compound.
 (1) Both Statement-I and Statement-II are correct.
 (2) Both Statement-I and Statement-II are incorrect.

- (3) Statement-I is correct & Statement-II is incorrect.
 (4) Statement-I is incorrect & Statement-II is correct.

Zoology : Section-A (Q. No. 151 to 185)

151. A characteristic of *Doliolum* is the presence of notochord:
 (1) only in larval tail.
 (2) from head to tail region.
 (3) throughout life.
 (4) only in adult tail.
152. Amino acids are substituted methanes. What are the four substituent groups occupying the four valency positions?
 (1) Hydrogen, carboxyl group, amino group and a variable group (R).
 (2) Two carboxyl groups, amino group and OH.
 (3) Two hydrogen, one carboxyl group, amino group and a variable group (R).
 (4) Two amino groups, one hydrogen and one carboxyl group.
153. Given below are two statements:
Statement I: The circulatory system of cockroaches contains a well developed aorta, vein and capillary.
Statement II: The blood of cockroaches does not contain any respiratory pigment.
 In the light of the above statements, choose the correct answer from the options given below:
 (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

154. Match List-I with List-II:

List-I	List-II
A. Green glands	I. Earthworms
B. Malpighian tubules	II. Flatworms
C. Protonephridia	III. Prawns
D. Nephridia	IV. Cockroaches

Choose the **correct** answer from the options given below:

- (1) A- IV, B-III, C-II, D-I
- (2) A-III, B-I, C-IV, D-II
- (3) A-II, B-III, C-I, D- IV
- (4) A-III, B- IV, C-II, D-I

155. Match List-I with List-II:

List-I	List-II
A. Tidal volume	I. 1000 - 1100 mL of air
B. Inspiratory reserve volume	II. 500mL of air
C. Expiratory reserve volume	III. 1100 - 1200 mL of air
D. Residual volume	IV. 2500 - 3000 mL of air

Choose the **correct** answer from the options given below:

- (1) A- II, B-I, C-IV, D-III
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-IV, C-I, D- III
- (4) A-III, B- II, C-IV, D-I

156. Choose the **correct** order for the pathway of conduction of action potential through the heart.

- (1) AV bundle → AV node → Purkinje fibres → SA node
- (2) SA node → AV node → AV bundle → Purkinje fibres
- (3) Purkinje fibres → SA node → AV bundle → AV node
- (4) AV node → SA node → AV bundle → Purkinje fibres

157. Which of the following pairs is **incorrectly** matched?

- (1) Hypersecretion of growth hormone- Acromegaly
- (2) Hyposecretion of vasopressin-Diabetes mellitus
- (3) Hypersecretion of thyroxine- Grave's disease
- (4) Normal secretion of melatonin- Maintains diurnal rhythm

158. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Proteins are homopolymers of amino acids.

Reason R: A homopolymer has only one type of monomer repeating 'n' number of times.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is NOT the correct explanation of A.

159. Match List-I with List-II:

List-I	List-II
A. Simple squamous epithelium	I. Bone
B. Simple cuboidal epithelium	II. Air sacs of lungs
C. Specialised connective tissue	III. Tubular parts of the nephron
D. Dense regular connective tissue	IV. Ligament

Choose the **correct** answer from the options given below:

- (1) A- II, B-III, C-I, D-IV
- (2) A-I, B-II, C-III, D-IV
- (3) A-III, B-IV, C-II, D- I
- (4) A-IV, B- II, C-I, D-III



160. Given below are some statements (A-D).
- Entamoeba histolytica* damages the small intestine of humans.
 - Wuchereria* infects or damages the upper limbs.
 - Trichophyton* is a bacterium that causes ringworm disease in humans.
 - Sporozoite after entering into human blood first infects the RBCs.

Choose the **incorrect** statements from the options given below.

- A only
- A and B only
- D only
- A, B, C and D

161. Given below are four statements (a)-(d) regarding embryonic development in humans.

- Cleavage divisions bring about considerable increase in the mass of protoplasm.
- With more cleavage divisions, the resultant blastomeres become smaller and smaller.
- The blastomeres in the blastocyst are arranged into two layers, trophoblast and endometrium.
- Cleavage division result in a solid ball of cells called morula.

Which of the above two statements are correct?

- (a) and (c)
- (b) and (d)
- (a) and (b)
- (c) and (d)

162. For the maintenance of ionic gradients across the resting membrane, the sodium-potassium pump transports;

- 3Na^+ outwards for 2K^+ into the cell.
- 2Na^+ outwards for 2K^+ into the cell.
- 2K^+ outwards for 3Na^+ into the cell.
- 2K^+ outwards for 2Na^+ into the cell.

163. Match the columns and choose the appropriate option.

Column-I		Column-II	
a.	GIFT	i.	Women who cannot produce ovum
b.	ZIFT	ii.	Introduction of the sperm directly into the ovum
c.	AI	iii.	Test tube baby
d.	ICSI	iv.	Men with low sperm count

- (a)-(i) (b)-(iii) (c)-(iv) (d)-(ii)
- (a)-(ii) (b)-(iii) (c)-(i) (d)-(iv)
- (a)-(iii) (b)-(ii) (c)-(i) (d)-(iv)
- (a)-(ii) (b)-(i) (c)-(iii) (d)-(iv)

164. A neural signal reaching the neuro-muscular junction releases a neurotransmitter (acetylcholine) which generates an action potential in:

- sarcolemma.
- sarcoplasmic reticulum.
- sarcoplasm.
- cross arm.

165. Match List-I with List-II:

List-I	List-II
A. Theory of panspermia	I. Survival of the fittest
B. Natural selection	II. Life came out of decaying and rotting matter.
C. Chemical evolution	III. Units of life called spores were transferred to different planets including Earth.
D. Theory of spontaneous generation	IV. Life could have come from pre-existing non-living organic molecules.

Choose the **correct** answer from the options given below:

- A- IV, B-III, C-II, D-I
- A-III, B-I, C-IV, D-II
- A-II, B-III, C-I, D- IV
- A-II, B-I, C-IV, D-III

166. AB blood group in humans is an example of:

- (1) incomplete dominance.
- (2) co-dominance.
- (3) polygenic inheritance.
- (4) pleiotropy.

167. Match List-I with List-II:

List-I	List-II
A. Concanavalin A	I. Drug
B. Vinblastin	II. Lectin
C. Anthocyanin	III. Toxin
D. Ricin	IV. Pigment

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-IV, B-I, C-II, D-III
- (3) A-II, B-III, C-I, D-IV
- (4) A-II, B-I, C-IV, D-III

168. Which of the following drugs slows down the body functions?

- (1) Heroin
- (2) Cocaine
- (3) Nicotine
- (4) Amphetamines

169. The following statements show some characteristic features of an organism. Identify the organism.

- A. It possesses longitudinal and circular muscles, which help in locomotion.
- B. A closed circulatory system is present.
- C. They are triploblastic, metamerically segmented and coelomate animals.
- D. It is a monoecious animal.
- E. Reproduction is sexual.

- (1) *Ancylostoma* (Hookworm)
- (2) *Hirudinaria*
- (3) *Chaetopleura*
- (4) *Clarias* (Magur)

170. What is the function of the vas deferens in the male reproductive system?

- (1) Production of testosterone
- (2) Maturation of sperm
- (3) Transport of sperm
- (4) Storage of sperm

171. Given below are two statements:

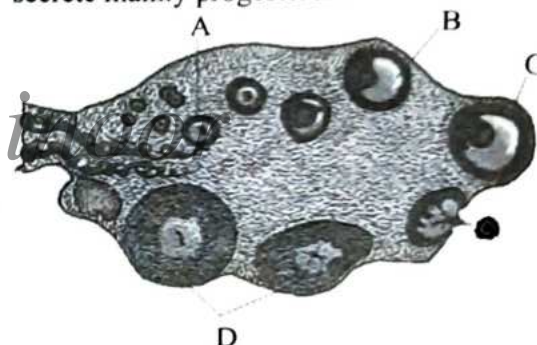
Statement I: Transgenic mice are being used to test the safety of the polio vaccine.

Statement II: Over 95% of all existing transgenic animals are mice.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

172. Which of the labelled parts in the given picture secrete mainly progesterone?



- (1) A
- (2) B
- (3) C
- (4) D

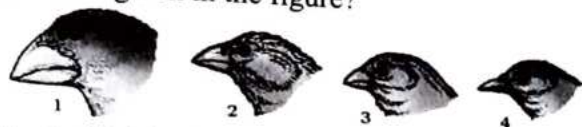
173. Given below are some statements (A- E).

- A. The junction between a motor neuron and the sarcolemma of the muscle fibre is called the motor end plate.
- B. Utilising the energy from ATP hydrolysis, the myosin head binds to the exposed active sites on actin to form a cross-bridge.
- C. The 'Z' line is attached to the thick filaments.
- D. The number of mitochondria are few in red muscle fibres.
- E. Red muscle fibres contain a red-coloured oxygen-storing pigment called myoglobin.

Choose the **correct** statements from the options given below.

- (1) A, B and C only (2) B, C and D only
(3) A, B and E only (4) C and D only

174. Which evolutionary phenomenon is depicted by the sketch given in the figure?



- (1) Artificial selection
(2) Genetic drift
(3) Convergent evolution
(4) Adaptive radiation
175. The brainstem consists of:
(1) pons and cerebrum.
(2) hypothalamus and cerebellum.
(3) medulla oblongata and cerebellum.
(4) medulla oblongata, midbrain and pons.
176. When there is a failure in the proper segregation of chromatids during cell division, it can lead to the gain or loss of chromosomes, resulting in an abnormal chromosomal number. This is called as:
(1) euploidy. (2) monoploidy.
(3) aneuploidy. (4) polyploidy.
177. Cofactor for the proteolytic enzyme carboxypeptidase is:
(1) zinc. (2) calcium.
(3) sodium. (4) magnesium.
178. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion A: Phenylpyruvic acid is excreted through urine in case of phenylketonuria.
Reason R: The affected individual lacks an enzyme that converts phenylalanine into tyrosine.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

179. Given below are some statements (A- D).

- A. Medical termination of pregnancy (MTP) during the first trimester is generally safe.
B. Generally, chances of conception are high until the mother breastfeeds the infant for up to two years.
C. Intrauterine devices like copper-T are effective contraceptives.
D. Contraceptive pills can be taken up to one week after coitus to prevent conception.

Choose the **correct** statements from the options given below:

- (1) A and B only
(2) C and D only
(3) A and C only
(4) B and D only

180. Match List-I with List-II:

List-I	List-II
A. Human immunodeficiency virus (HIV)	I. Amoebiasis
B. <i>Wuchereria bancrofti</i>	II. AIDS
C. <i>Entamoeba histolytica</i>	III. Filariasis
D. <i>Plasmodium falciparum</i>	IV. Malaria

Choose the **correct** answer from the options given below:

- (1) A- II, B-III, C-IV, D-I
(2) A-I, B-III, C-IV, D-II
(3) A-II, B-III, C-I, D- IV
(4) A-II, B- IV, C-I, D-III

181. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Very small animals are rarely found in polar areas.

Reason R: Small animals have a larger surface area relative to their volume.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

182. What is the **correct** order (old to recent) of periods in the Paleozoic era?

- (1) Silurian → Devonian → Permian → Carboniferous
- (2) Silurian → Devonian → Carboniferous → Permian
- (3) Permian → Devonian → Silurian → Carboniferous
- (4) Silurian → Carboniferous → Permian → Devonian

183. Match List-I with List-II:

List-I	List-II
A. Gametogenesis	I. Transfer of sperms into the female genital tract
B. Insemination	II. Delivery of the baby
C. Implantation	III. Formation of gametes
D. Parturition	IV. Attachment of blastocyst with endometrium

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-III, B-I, C-IV, D-II
- (3) A-IV, B-I, C-II, D-III
- (4) A-II, B-IV, C-I, D-III

184. Which of the following are **not** strictly biomacromolecules but present in an acid-insoluble fraction?

- (1) Nucleic acids
- (2) Lipids
- (3) Amino acid
- (4) Simple sugars

185. Females with Turner's syndrome have:

- (1) round head.
- (2) gynaecomastia.
- (3) furrowed tongue.
- (4) rudimentary ovaries.

Zoology : Section-B (Q. No. 186 to 200)

186. Match the List-I with List-II:

List-I (Scientific names)	List-II (Common names)
A. <i>Physalia</i>	I. Liver fluke
B. <i>Taenia</i>	II. Scypha
C. <i>Fasciola</i>	III. Tapeworm
D. <i>Sycon</i>	IV. Portuguese man of war

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-III, D-I
- (2) A-IV, B-III, C-I, D-II
- (3) A-I, B-III, C-II, D-IV
- (4) A-I, B-II, C-III, D-IV

187. Graft rejection can be prevented by:

- (1) antibodies.
- (2) T cells.
- (3) immunostimulatory drugs.
- (4) immunosuppressant drugs.

188. Which of the following are essential processes for defining living organisms?

- (a) Metabolism
- (b) Reproduction
- (c) Growth
- (d) Ability to sense the environment
- (e) Cellular organization

Choose the correct group of processes from the options given below:

- (1) (a), (b), and (c) only
- (2) (a), (d) and (e) only
- (3) (b), (c), and (d) only
- (4) (a), (c), and (e) only

189. Given below are two statements:

Statement I: 'Mac Arthur' showed that five closely related species of warblers living on the same tree able to avoid competition and co-exist due to behavioural differences in their foraging activities.

Statement II: The life cycles of parasites are often complex, involving one or two intermediate hosts or vectors to facilitate parasitisation of its primary host.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

190. Choose the **incorrect** statement about kidneys.

- (1) Glomerulus is a tuft of capillaries formed by a fine branch of the renal vein.
- (2) Bowman's capsule along with the glomerulus is called the renal corpuscle.
- (3) Each kidney has nearly 1 million complex tubular structures, called nephrons.
- (4) The glomerulus is not a part of the renal tubule.

191. Given below are some statements (A- D).

- A. Male accessory glands include a pair of seminal vesicles, a paired bulbourethral glands and a prostate gland.
- B. Secretion of bulbourethral glands helps in the formation of sperms.
- C. Sperm formation ceases at the age of 50 years in men.
- D. The period of embryonic development is called gestation.

Choose the **incorrect** statements from the options given below:

- (1) A and B only
- (2) B and C only
- (3) A and D only
- (4) B and D only

192. Match the **List-I** with **List-II**:

List-I	List-II
A. Facial bones	I. 14
B. Bones in the axial skeleton	II. 8
C. Cranial bones	III. 80
D. Bones in the vertebral column	IV. 26

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-IV, B-III, C-I, D-II
- (3) A-I, B-III, C-II, D-IV
- (4) A-I, B-II, C-III, D-IV

193. Given below are some statements (A- D).

- A. Analogous structures are a result of divergent evolution.
- B. The first mammals were like shrews.
- C. About 65 mya, the dinosaurs suddenly disappeared from the earth.
- D. About 200 million years ago (mya), the first cellular forms of life appeared on Earth.

Choose the **correct** statements from the options given below:

- (1) A and B only
- (2) B and C only
- (3) A and D only
- (4) B and D only

194. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Androgens stimulate muscular growth.

Reason R: Androgens produce catabolic effects on protein and carbohydrate metabolism.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is NOT the correct explanation of A.

195. Given below are two statements:

Statement I: The neural system is composed of highly specialised cells called nephrons that can detect stimuli.

Statement II: Association areas in cerebral cortex are neither clearly sensory nor motor in function.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

196. Read the following statements and choose correct one.

- (1) RBC have an average life span of 120 days after which all these are destroyed in liver.
- (2) Basophils secrete histamine, serotonin, heparin, etc. and are involved in inflammatory reactions.
- (3) Leucocytes are non-nucleated cells and are relatively lesser in number.
- (4) Eosinophils are most abundant among granulocytes and associated with bacterial infections.

197. CDRI stands for:

- (1) Contraceptive Drug Research Institute.
- (2) Central District Research Institute.
- (3) Central Drug Research Institute.
- (4) Central Dermatologic Research Institute.

198. Inspiration occurs when intra-pulmonary pressure is:

- (1) higher than atmospheric pressure.
- (2) lower than atmospheric pressure.
- (3) equal to atmospheric pressure.
- (4) zero compared to atmospheric pressure.

199. Given below are two statements:

Statement I: Primers are small chemically synthesised oligonucleotides that are complementary to the specific regions of DNA.

Statement II: If any protein-encoding gene is expressed in a heterologous host, it is called a recombinant protein.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

200. In the digestive system of cockroaches, gastric caeca is present at the junction of:

- (1) midgut and hindgut.
- (2) hindgut and foregut.
- (3) foregut and mouth.
- (4) foregut and midgut.

■■■